Births in North Carolina

4/28/2025

This is an individual assignment. You may discuss it in general terms with others from our class, but you should be doing the work on your own. As always, of course, is not appropriate to copy the work of any other students. You are welcome to ask me questions as you work and to use other resources, including things I have posted on Blackboard, our textbook, and code you look up online. If you are using substantial code from online, please mention your source.

In this project, you will use your R skills to examine and predict birth weights of babies.

Getting set up and exploring the data.

1. Load the packages you may need: `tidyverse`, `broom`, `scales`, and `ggdist`.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(broom)
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```
library(ggdist)
```

2. Next, we will load the dataset we are going to use. The dataset has been provided by our textbook authors and is located at https://www.openintro.org/data/csv/ncbirths.csv

```
ncbirths <- read_csv("data/ncbirths.csv")
```

```
## Rows: 1000 Columns: 13
```

```
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (7): mature, premie, marital, lowbirthweight, gender, habit, whitemom
## dbl (6): fage, mage, weeks, visits, gained, weight
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

3. Next, peruse the data. View the dataframe in the usual ways. Also examine the codebook for the data which is located online at https://www.openintro.org/data/index.php?data=ncbirths. It is important that you have correct interpretations of the variables you will be discussing. Then, answer the following questions about the data.

```r
#explore data + brief summary stats
head(ncbirths, 10)
```

```
## # A tibble: 10 x 13
##     fage  mage mature   weeks premie visits marital gained weight lowbirthweight
##    <dbl> <dbl> <chr>    <dbl> <chr>   <dbl> <chr>    <dbl>  <dbl> <chr>
##  1    NA    13 younger~    39 full ~     10 not ma~     38   7.63 not low
##  2    NA    14 younger~    42 full ~     15 not ma~     20   7.88 not low
##  3    19    15 younger~    37 full ~     11 not ma~     38   6.63 not low
##  4    21    15 younger~    41 full ~      6 not ma~     34   8    not low
##  5    NA    15 younger~    39 full ~      9 not ma~     27   6.38 not low
##  6    NA    15 younger~    38 full ~     19 not ma~     22   5.38 low
##  7    18    15 younger~    37 full ~     12 not ma~     76   8.44 not low
##  8    17    15 younger~    35 premie      5 not ma~     15   4.69 low
##  9    NA    16 younger~    38 full ~      9 not ma~     NA   8.81 not low
## 10    20    16 younger~    37 full ~     13 not ma~     52   6.94 not low
## # i 3 more variables: gender <chr>, habit <chr>, whitemom <chr>
```

```r
summary(ncbirths)
```

```
##       fage            mage        mature               weeks
##  Min.   :14.00   Min.   :13   Length:1000        Min.   :20.00
##  1st Qu.:25.00   1st Qu.:22   Class :character   1st Qu.:37.00
##  Median :30.00   Median :27   Mode  :character   Median :39.00
##  Mean   :30.26   Mean   :27                      Mean   :38.33
##  3rd Qu.:35.00   3rd Qu.:32                      3rd Qu.:40.00
##  Max.   :55.00   Max.   :50                      Max.   :45.00
##  NA's   :171                                     NA's   :2
##     premie              visits       marital              gained
##  Length:1000        Min.   : 0.0   Length:1000        Min.   : 0.00
##  Class :character   1st Qu.:10.0   Class :character   1st Qu.:20.00
##  Mode  :character   Median :12.0   Mode  :character   Median :30.00
##                     Mean   :12.1                      Mean   :30.33
##                     3rd Qu.:15.0                      3rd Qu.:38.00
##                     Max.   :30.0                      Max.   :85.00
##                     NA's   :9                         NA's   :27
##      weight       lowbirthweight        gender              habit
##  Min.   : 1.000   Length:1000        Length:1000        Length:1000
##  1st Qu.: 6.380   Class :character   Class :character   Class :character
##  Median : 7.310   Mode  :character   Mode  :character   Mode  :character
##  Mean   : 7.101
##  3rd Qu.: 8.060
```

```
##   Max.   :11.750
##
##    whitemom
##   Length:1000
##   Class :character
##   Mode  :character
##
##
##
##
```

4. From your exploration of the dataframe and the codebook, was this study a randomized experiment or an observational study? How can you tell?

This was an observational study because the researchers did not manipulate any variables. They only observed and recorded information about births without assigning treatments to the mothers.

5. Will we be able to draw conclusions about causation from the work we do with this dataset? Explain.

No, we cannot draw conclusions about causation because this is an observational study, not a randomized experiment. We can only find associations between variables.

6. From your exploration of the dataframe and the codebook, can we generalize the results of this study to some larger population, do you think? Explain carefully.

Yes, since the data is a random sample of births from North Carolina, the results can be generalized to births in North Carolina around the time the data was collected. However, other factors such as differences in healthcare and demographics should be taken into consideration assuming we are trying to generalize to births everywhere.

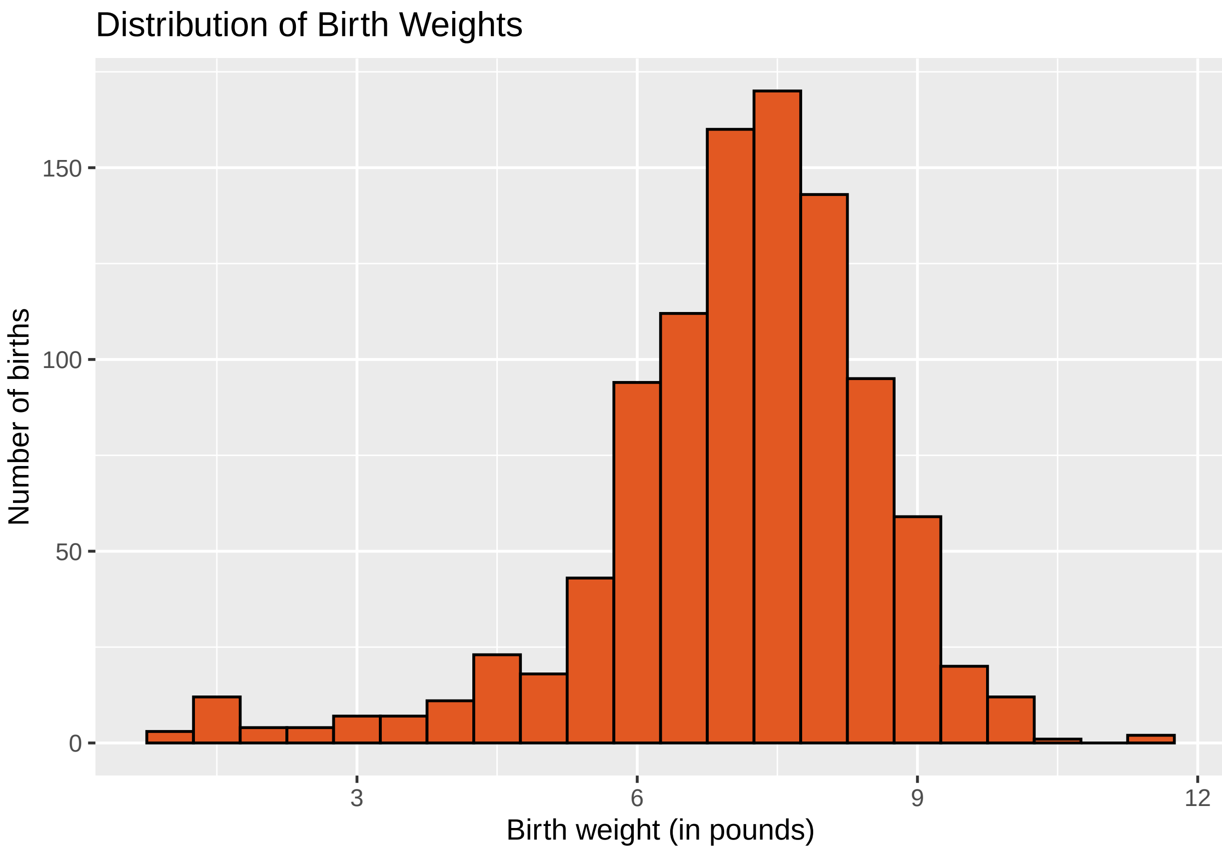
7. Examine the variables `weight`, `gender`, and `weeks`. Copy their meanings from the codebook. Classify each one as categorical or numerical.

'weight': Weight of the baby at birth in pounds. (numerical) 'gender': Gender of the baby, female or male. (categorical) 'weeks': Length of pregnancy in weeks. (numerical)

A closer look at the variable `weight`.

8. Create an appropriate graph for the distribution of the single variable `weight`. One of the next questions will ask you to describe the distribution so make sure that you are choosing a graph that makes this job easy to do. (Hint: What type of variable is this? What sorts of graphs are appropriate for a variable of this type?) Add a title and labels.

```r
# hist of births using weight
ggplot(data = ncbirths, mapping = aes(x = weight)) +
  geom_histogram(binwidth = 0.5, fill = "#E25822", color = "black") +
  labs(title = "Distribution of Birth Weights",
       x = "Birth weight (in pounds)",
       y = "Number of births")
```



9. Calculate appropriate summary statistics for the variable `weight`.

```
#summary stats
ncbirths %>%
  summarize(
    mean_weight = mean(weight, na.rm = TRUE),
    median_weight = median(weight, na.rm = TRUE),
    sd_weight = sd(weight, na.rm = TRUE),
    min_weight = min(weight, na.rm = TRUE),
    max_weight = max(weight, na.rm = TRUE))
```

```
## # A tibble: 1 x 5
##   mean_weight median_weight sd_weight min_weight max_weight
##         <dbl>         <dbl>     <dbl>      <dbl>      <dbl>
## 1        7.10          7.31      1.51          1       11.8
```

10. Describe the distribution of the variable `weight`, highlighting the key features of the distribution, as we have been practicing, and speaking in context. Make sure to support what you say with the statistics you have calculated.

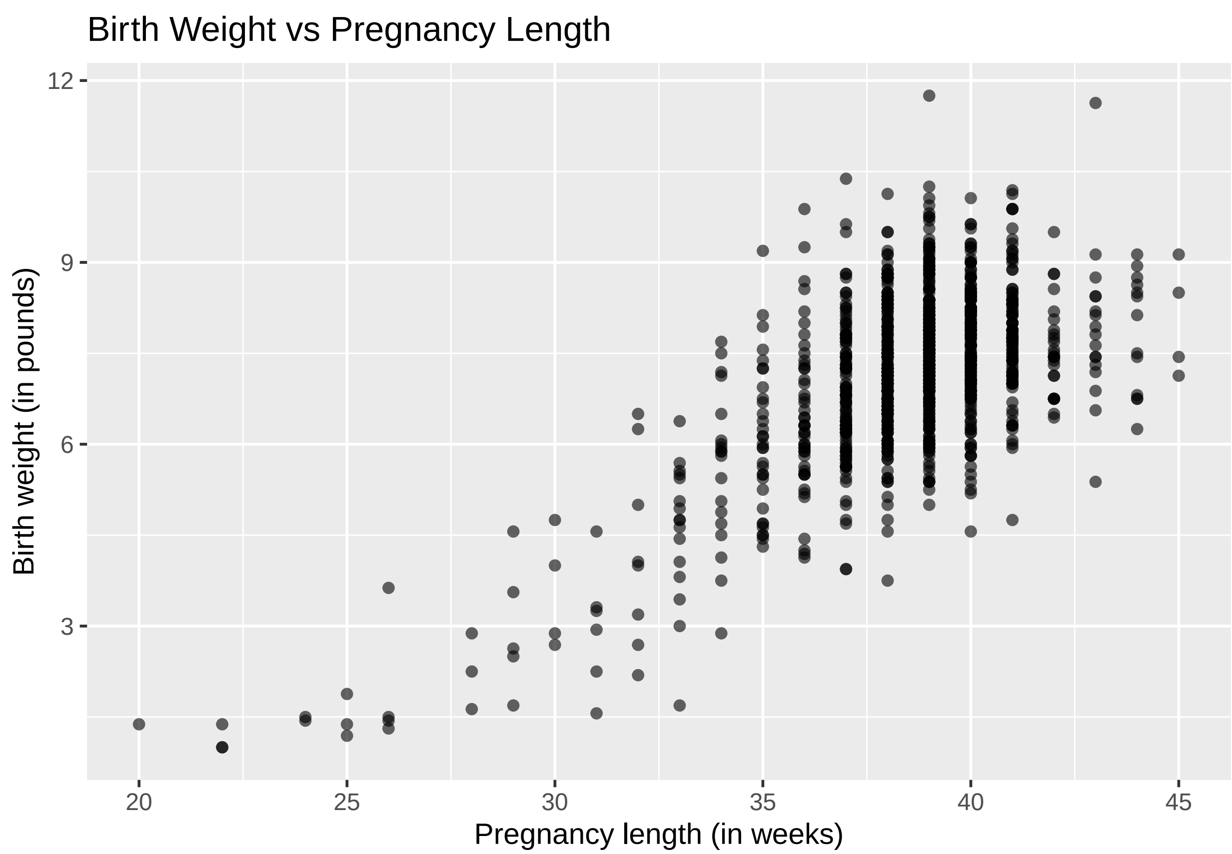
The distribution of birth weights is bell shaped, with most weights falling between about 6 and 8 pounds. The mean birth weight is 7.101 pounds, and the median is slightly higher at 7.31 pounds, suggesting a very slight left skew, but overall the shape is roughly normal. The standard deviation is about 1.5109 pounds, indicating moderate spread.Then the minimum weight is at 1 pound, and the maximum is 11.75 pounds.

Predicting weight

11. I am interested to see how the length of a mother's pregnancy impacts birthweight. Make an appropriate scatterplot as a first step for investigating this question.

```
ggplot(data = ncbirths, aes(x = weeks, y = weight)) +
  geom_point(alpha = 0.6) +
  labs(title = "Birth Weight vs Pregnancy Length",
       x = "Pregnancy length (in weeks)",
       y = "Birth weight (in pounds)")
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



12. Use `lm` to create a linear model to answer the research question. Print the model output summary.

```
weight_lm <- lm(weight ~ weeks, data = ncbirths)
summary(weight_lm)
```

```
##
## Call:
## lm(formula = weight ~ weeks, data = ncbirths)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.5775 -0.7048 -0.0235  0.7022  4.4165
```

```
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6.09529    0.46464  -13.12   <2e-16 ***
## weeks        0.34433    0.01209   28.49   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.119 on 996 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.449,  Adjusted R-squared:  0.4485
## F-statistic: 811.7 on 1 and 996 DF,  p-value: < 2.2e-16
```

13. Write the equation of the model using appropriate notation.

weight_hat = -6.095 + 0.344 * weeks

note: Where 'weeks' is the length of pregnancy (in weeks)

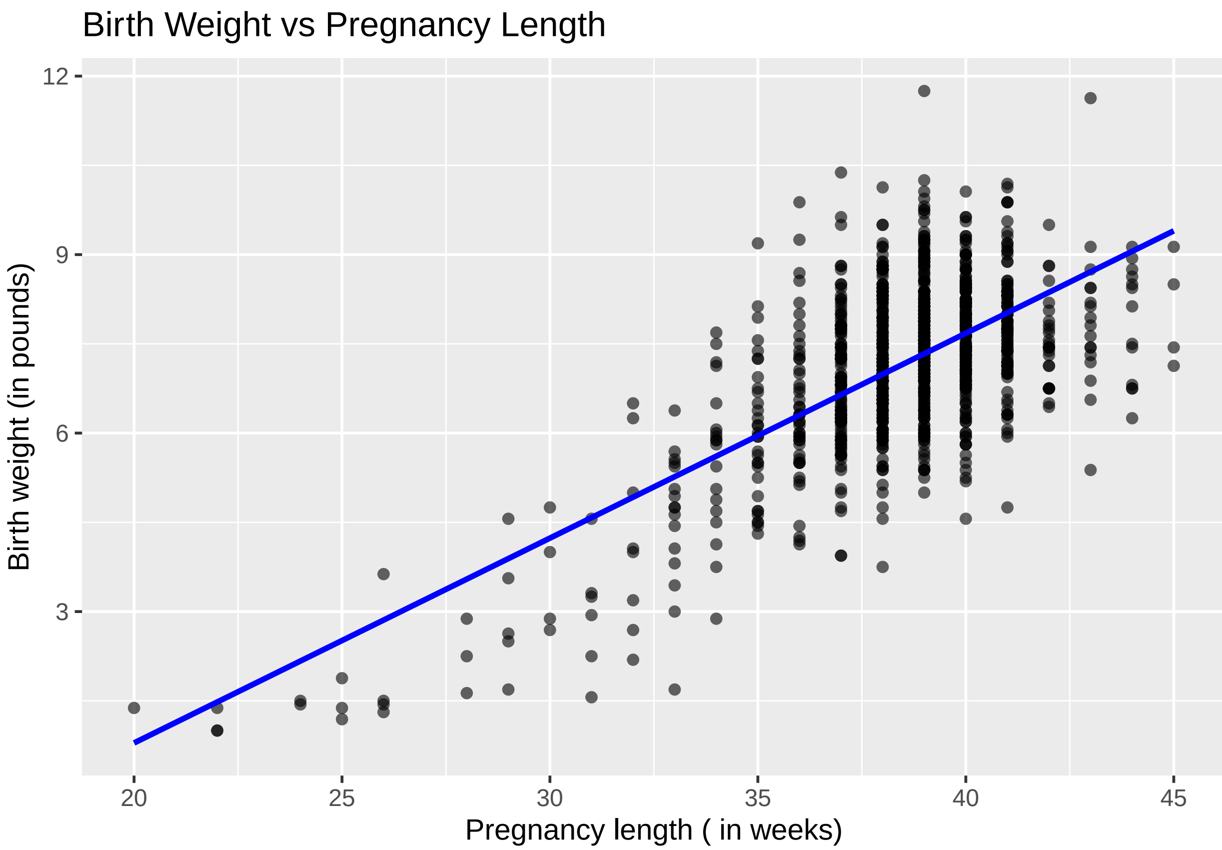
14. Graph the original data together with the model. Add a title and labels to your graph. Address any overplotting issues.

```r
# scatterplot of data + regression line from the model
ggplot(data = ncbirths, aes(x = weeks, y = weight)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(title = "Birth Weight vs Pregnancy Length",
       x = "Pregnancy length ( in weeks)",
       y = "Birth weight (in pounds)")
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



15. Write a paragraph to describe the relationship between the birthweight of a baby and the length of the mother's pregancy, as you should be seeing in your scatterplot. Make sure to address the key features of scatterplots and to always speak in context, supporting what you say with numbers.

The scatter plot displays data from births recorded in North Carolina. There is a moderate positive linear association between pregnancy length and birth weight. In general, as pregnancy length increases, birth weight also tends to increase. Most babies born around 36 weeks have a birth weight close to 6.1 pounds, while babies born around 40 weeks tend to have birth weights close to 7.5 to 8 pounds. However, there are exceptions to this trend. For instance, some babies born at 36 weeks weigh more than 7 pounds, while some full-term babies (around 40 weeks give or take) have birth weights under 6 pounds. In summary, the trend suggests that longer pregnancies are associated with heavier birth weights.

16. Write a sentence of interpretation for the slope of your model, in context.

For each additional week of pregnancy, the baby's birth weight is expected to increase by about 0.344 pounds.

17. What does your model predict for the birthweight of a baby who is born at 36 weeks?

```r
# Predict birthweight for a baby born at 36 weeks
predicted_weight <- -6.09529 + (0.34433 * 36)
predicted_weight
```

```
## [1] 6.30059
```

18. There is a baby in the dataset who weighed 7.25lbs when born at 36 weeks. Calculate the residual for this baby, showing your work.

```r
# only need actual weight as the predicted is already a variable from above chunk
actual_weight <- 7.25

# calculate residual
residual_36 <- actual_weight - predicted_weight
residual_36
```

```
## [1] 0.94941
```